

9 (a) (i) non-inverting (amplifier) B1 [1]

(ii) $(G =) 1 + R_2 / R_1$ B1 [1]

(b) (i) gain = $1 + 100 / 820$ C1
output = 17 mV A1 [2]

(ii) 9V A1 [1]

(R_2 / R_1) scores 0 in (a)(ii) but possible 1 mark in each of (b)(i) and (b)(ii)

$(1 + R_1 / R_2)$ scores 0 in (a)(ii), no mark in (b)(i), possible 1 mark in (b)(ii)

$(1 - R_2 / R_1)$ or R_1 / R_2 scores 0 in (a)(ii), (b)(i) and (b)(ii)

10 (a) (i) density $\times$ speed of wave (in the medium) B1 [1]